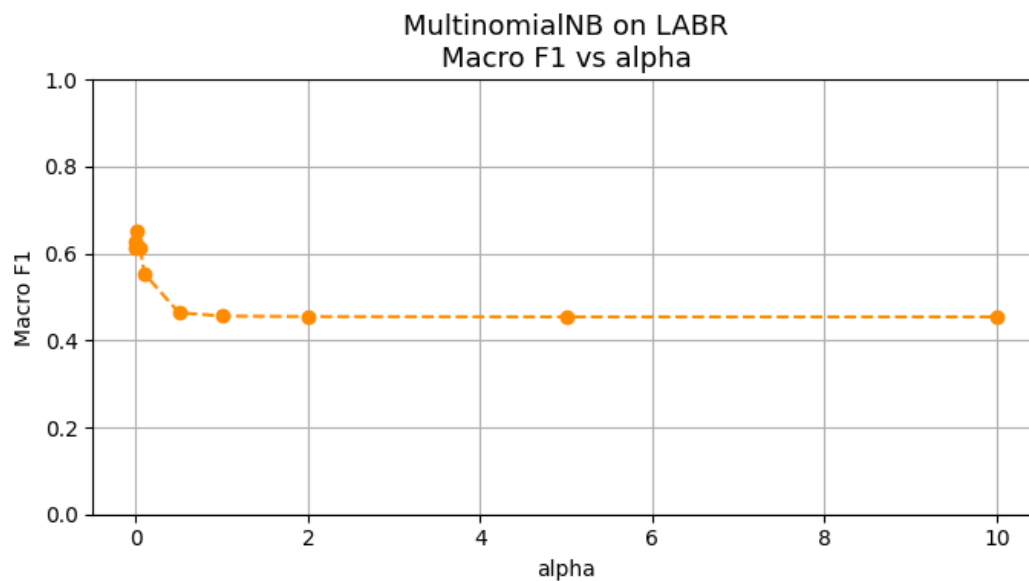
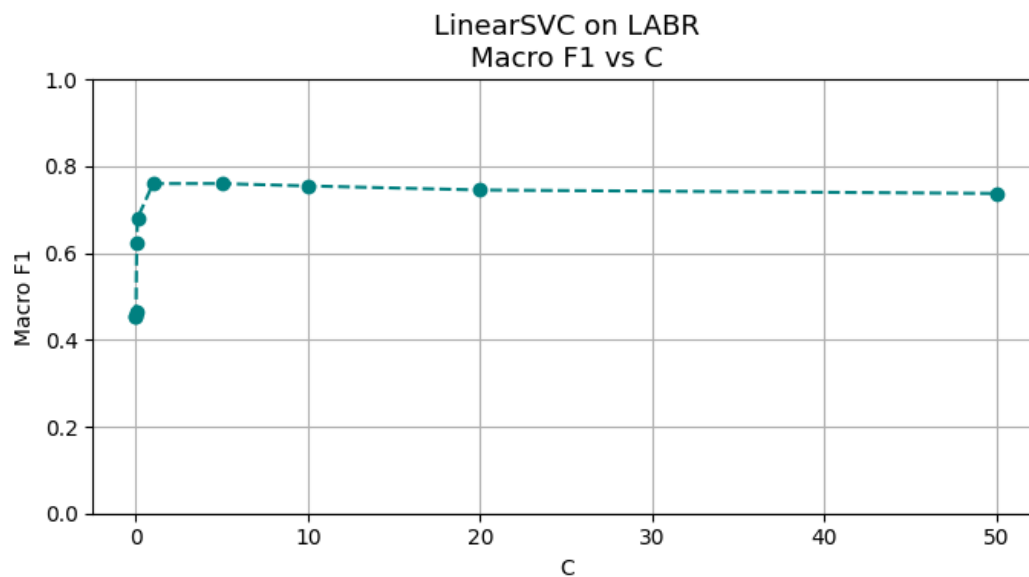


Analyse de l'effet des hyperparamètres pour chaque algorithme

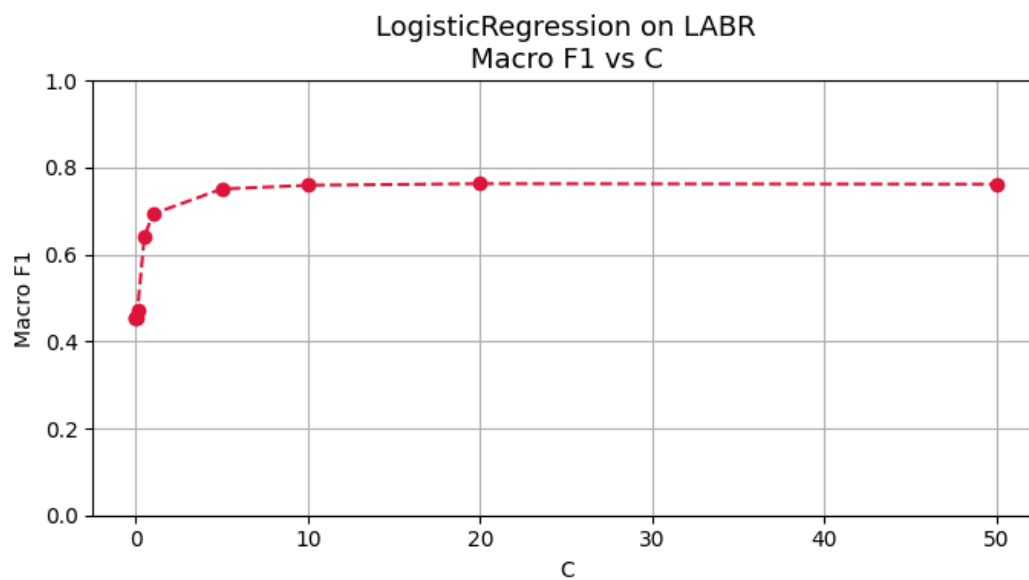
Multinomial Naïve Bayes (alpha)



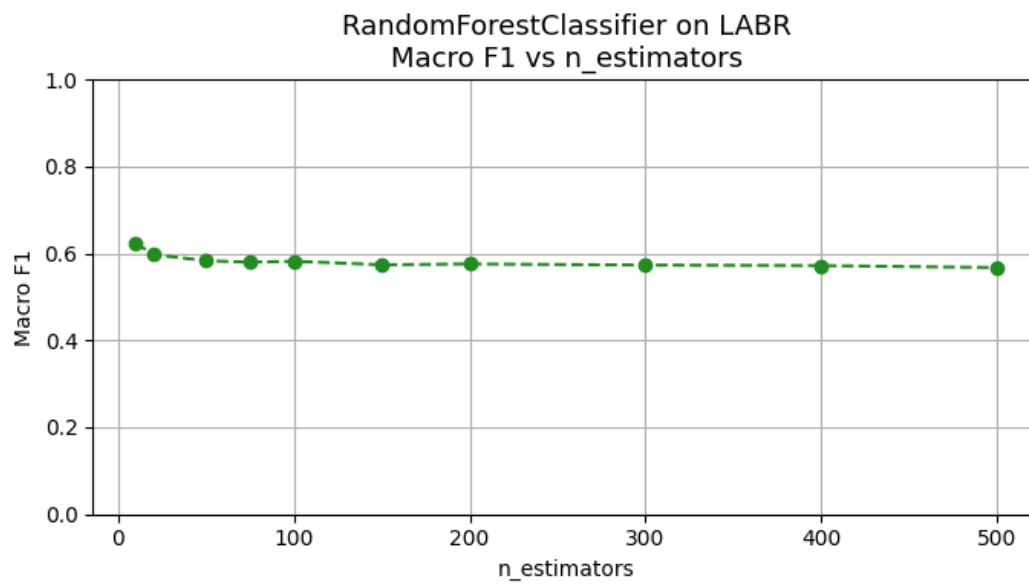
SVM-Linear (C)



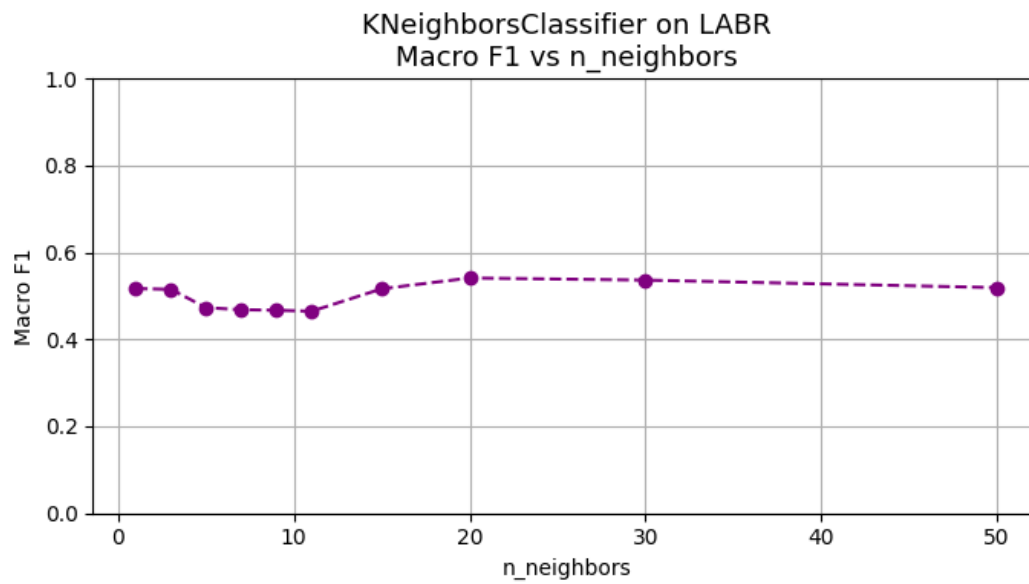
Logistic Regression (C)



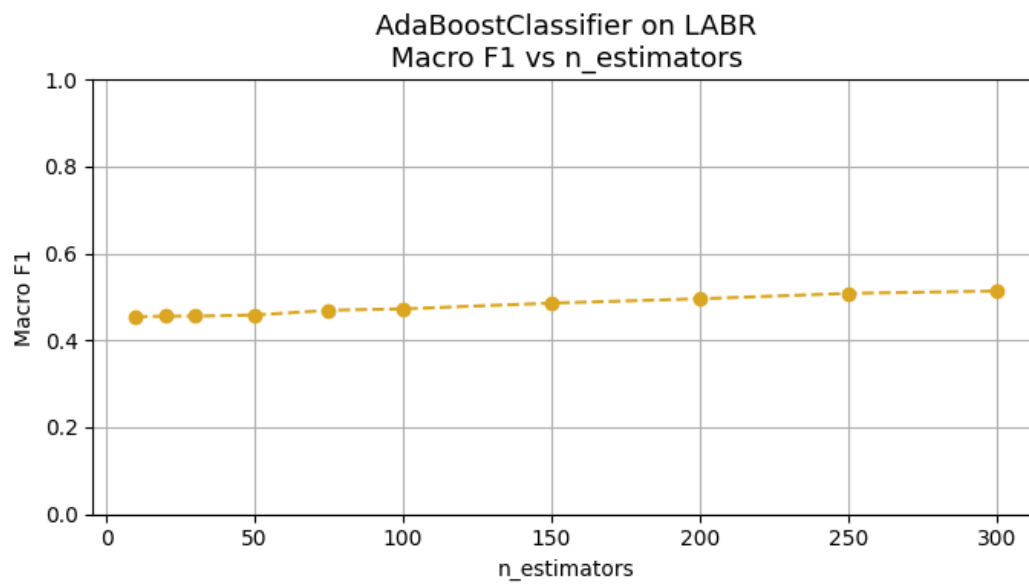
Random Forest (n_estimators)



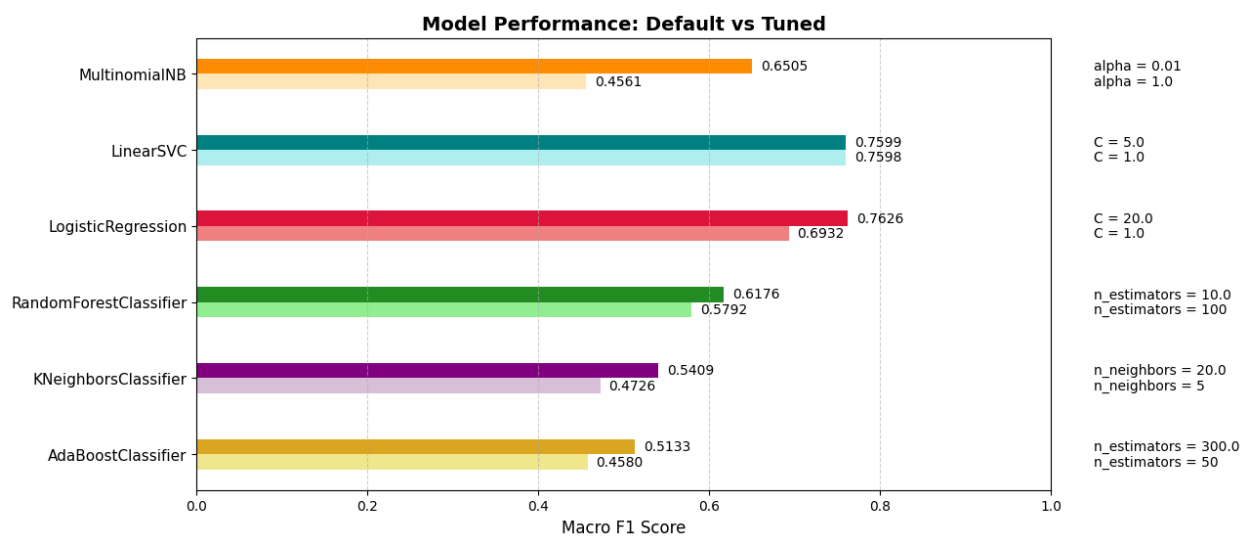
K-Nearest Neighbors (n_neighbors)



AdaBoost Classifier (n_estimators)



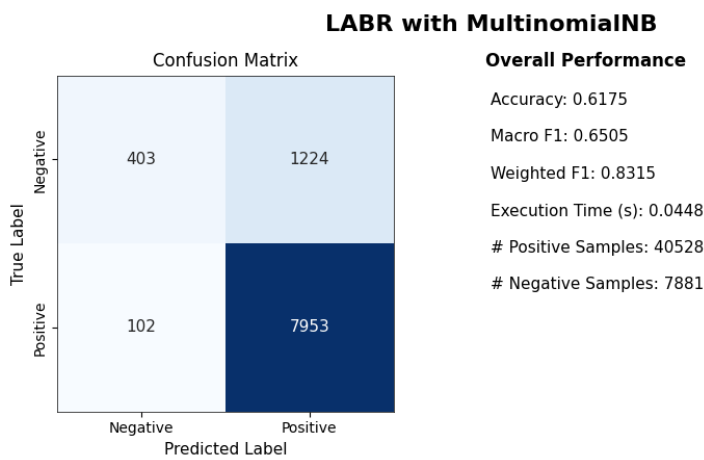
Comparaison des performances: Hyperparamètres par défaut vs Optimisés



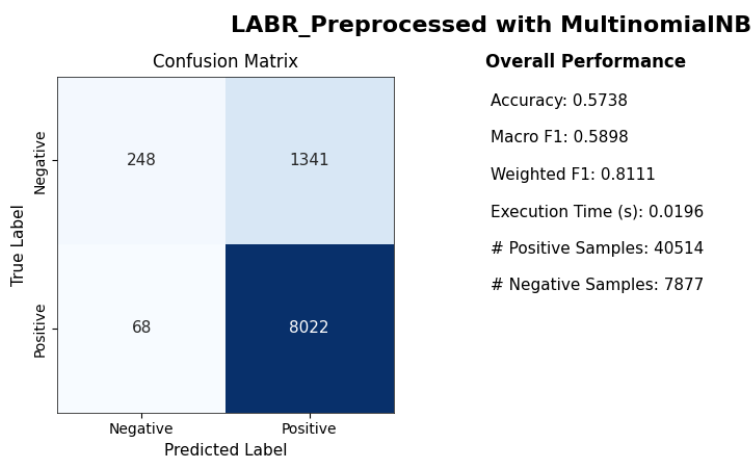
Évaluation comparative sur différentes versions du dataset

Multinomial Naïve Bayes

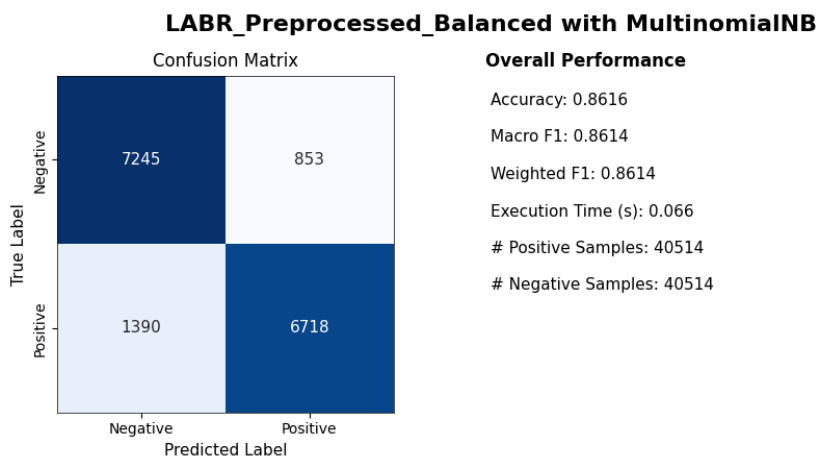
LABR (Brut)



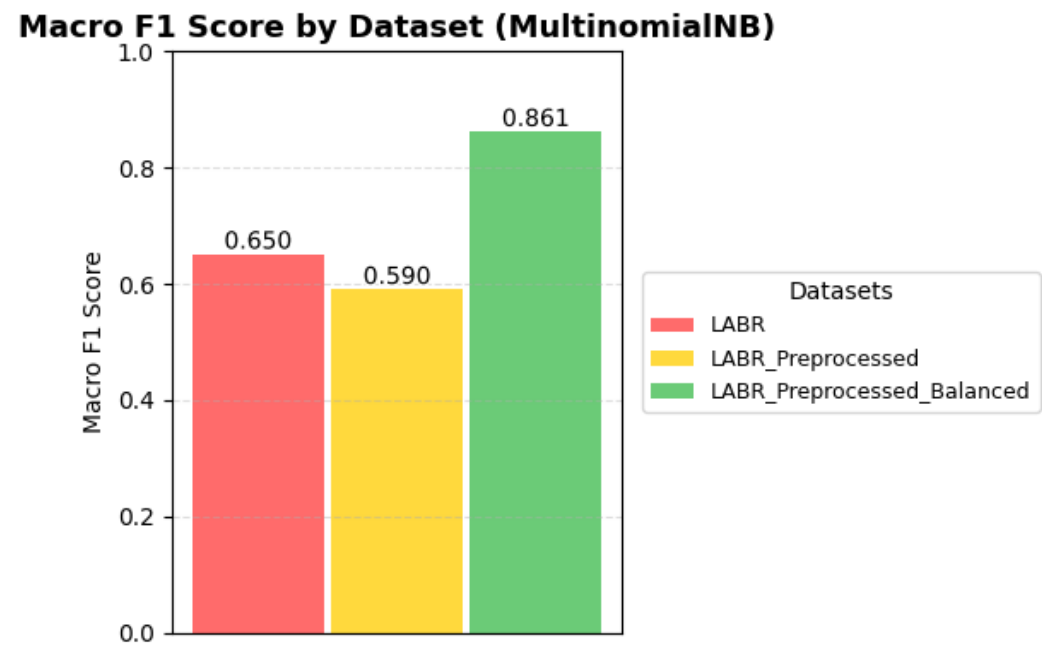
LABR (Prétraité)



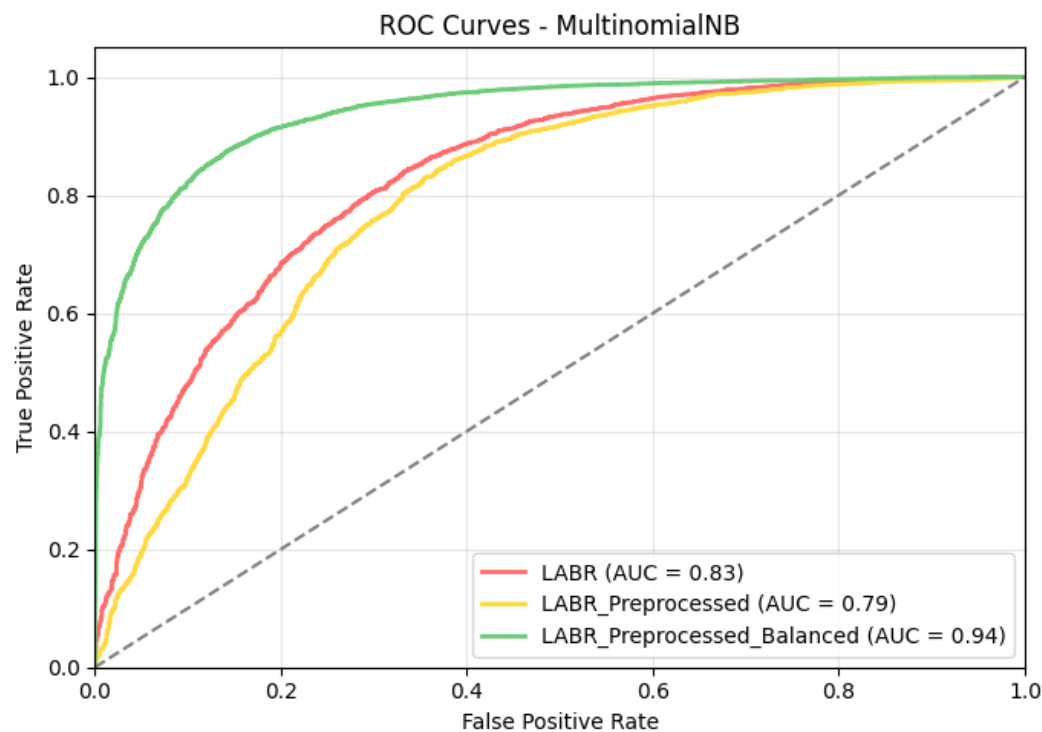
LABR (Prétraité + Équilibré)



Comparaison: Macro F1

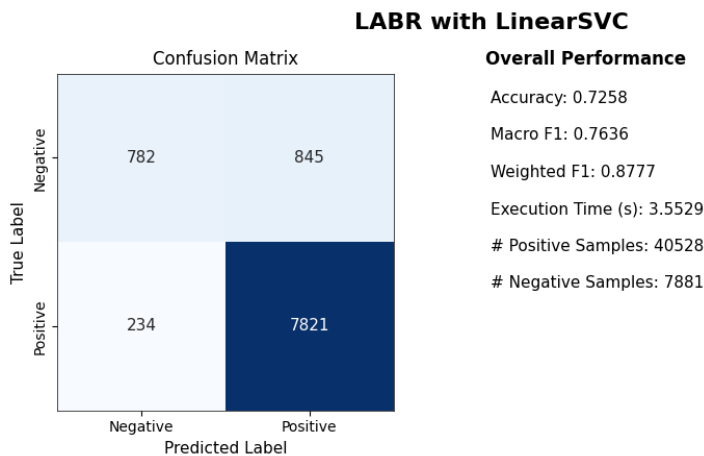


Comparison: ROC



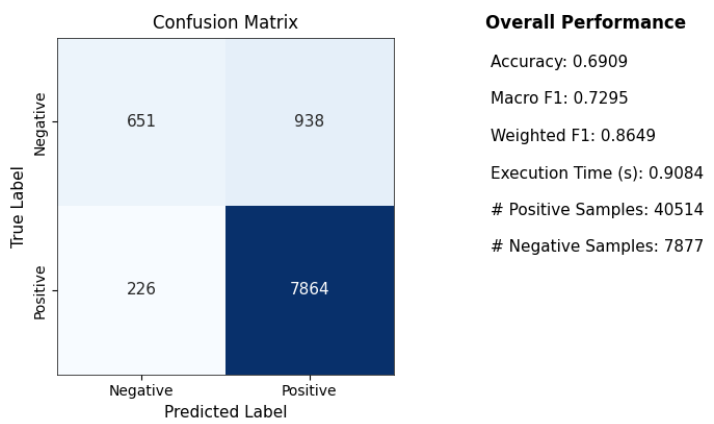
SVM-Linear

LABR (Brut)



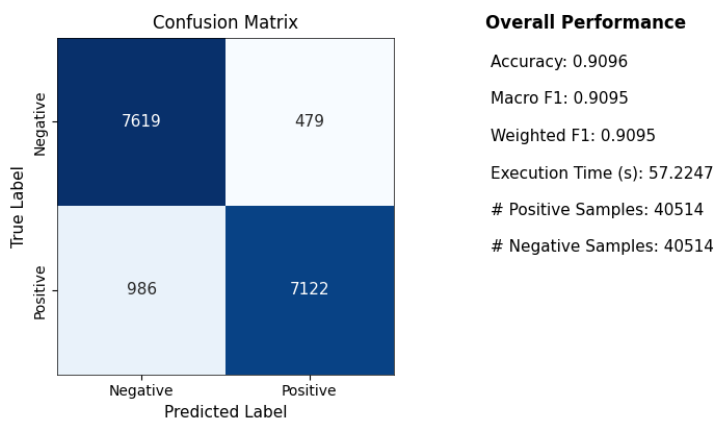
LABR (Prétraité)

LABR_Preprocessed with LinearSVC



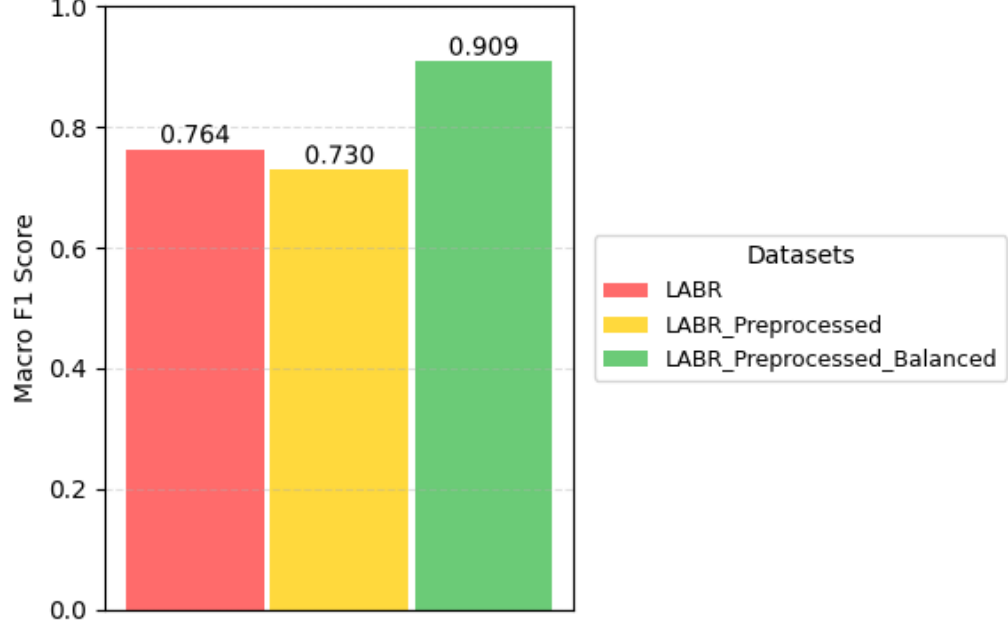
LABR (Prétraité + Équilibré)

LABR_Preprocessed_Balanced with LinearSVC

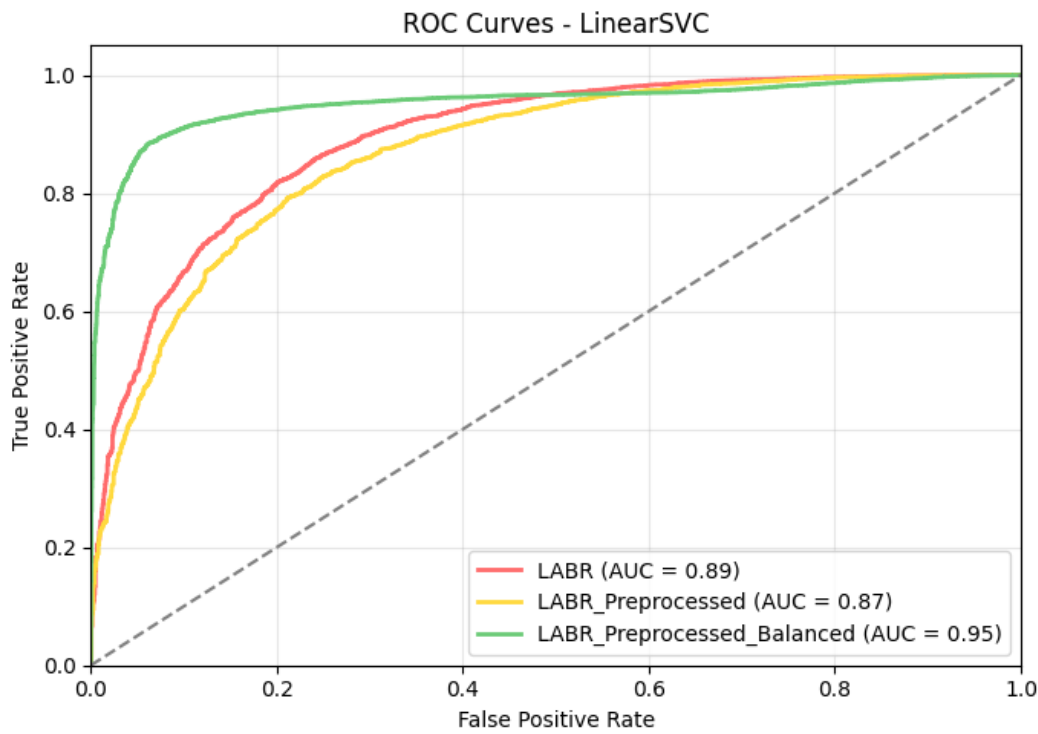


Comparaison: Macro F1

Macro F1 Score by Dataset (LinearSVC)



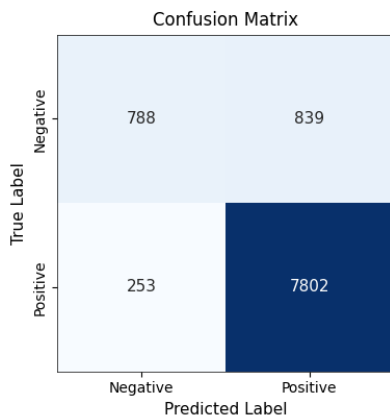
Comparison: ROC



Logistic Regression

LABR (Brut)

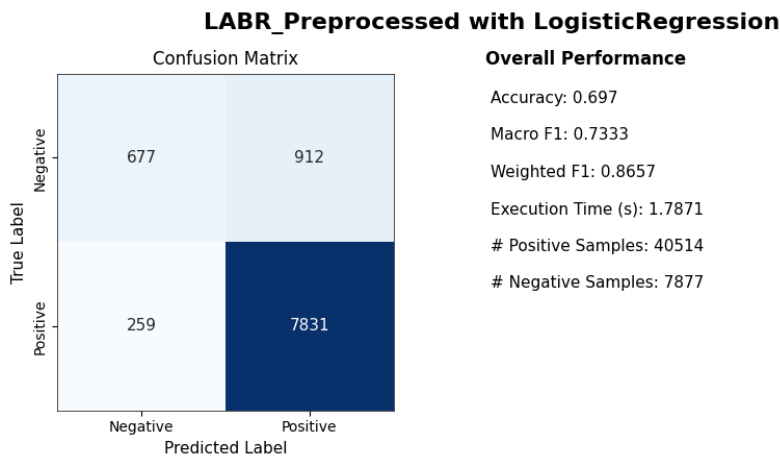
LABR with LogisticRegression



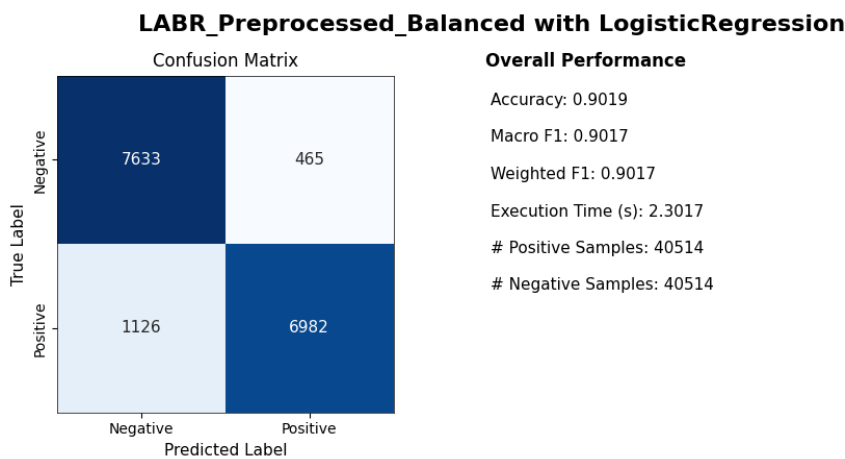
Overall Performance

Accuracy: 0.7265
Macro F1: 0.7626
Weighted F1: 0.8768
Execution Time (s): 8.0214
Positive Samples: 40528
Negative Samples: 7881

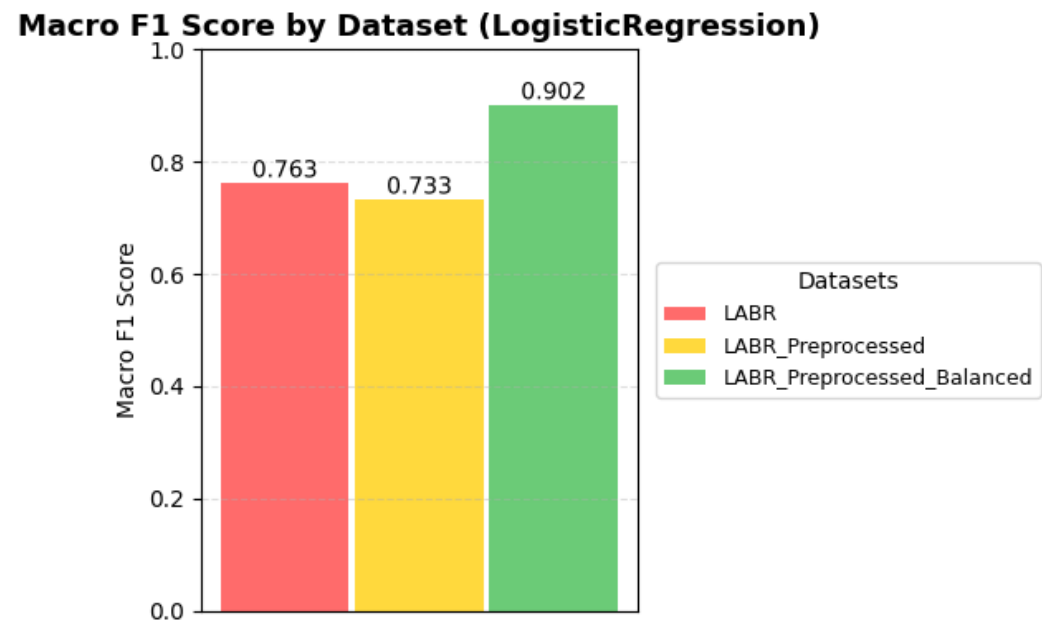
LABR (Prétraité)



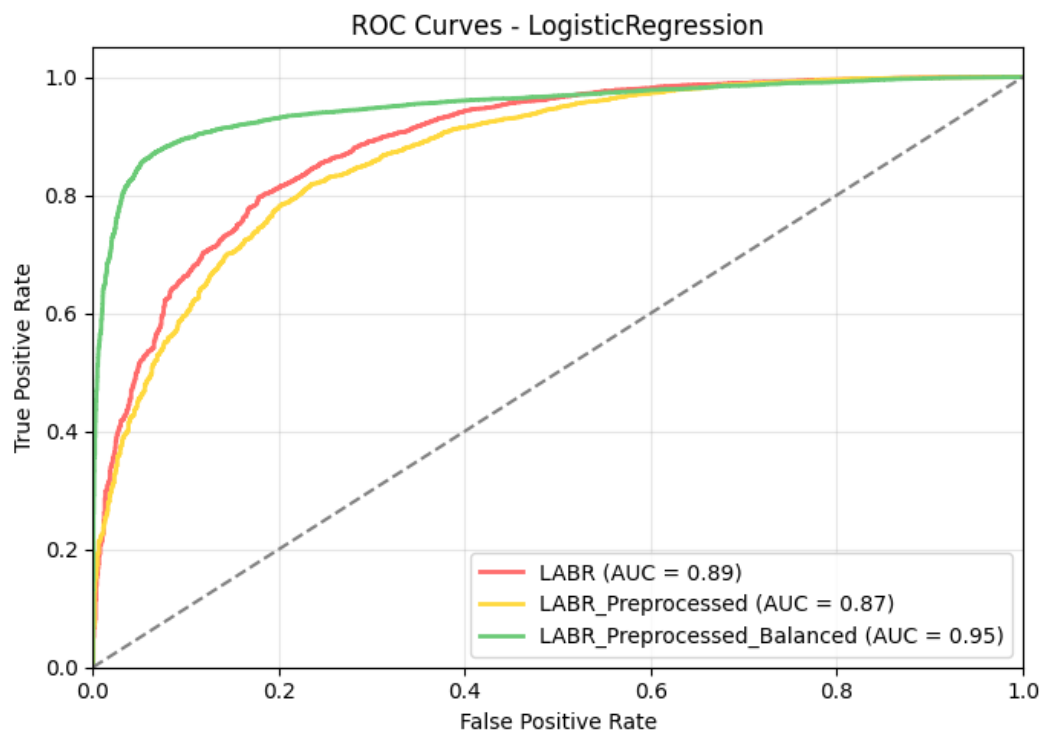
LABR (Prétraité + Équilibré)



Comparaison: Macro F1



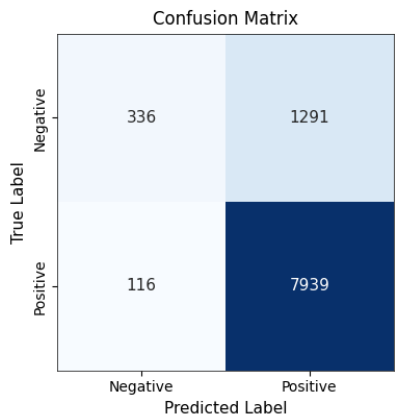
Comparison: ROC



Random Forest

LABR (Brut)

LABR with RandomForestClassifier



Overall Performance

Accuracy: 0.5961

Macro F1: 0.6209

Weighted F1: 0.8186

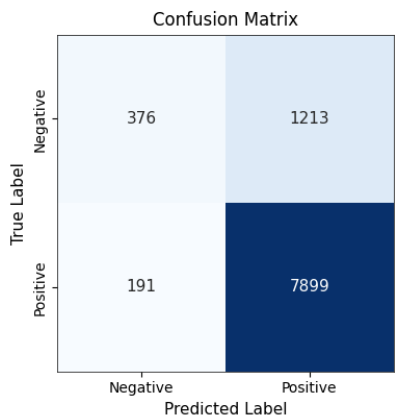
Execution Time (s): 30.6461

Positive Samples: 40528

Negative Samples: 7881

LABR (Prétraité)

LABR_Preprocessed with RandomForestClassifier



Overall Performance

Accuracy: 0.6065

Macro F1: 0.6336

Weighted F1: 0.8249

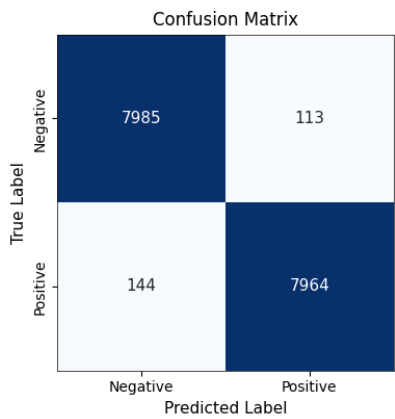
Execution Time (s): 12.681

Positive Samples: 40514

Negative Samples: 7877

LABR (Prétraité + Équilibré)

LABR_Preprocessed_Balanced with RandomForestClassifier



Overall Performance

Accuracy: 0.9841

Macro F1: 0.9841

Weighted F1: 0.9841

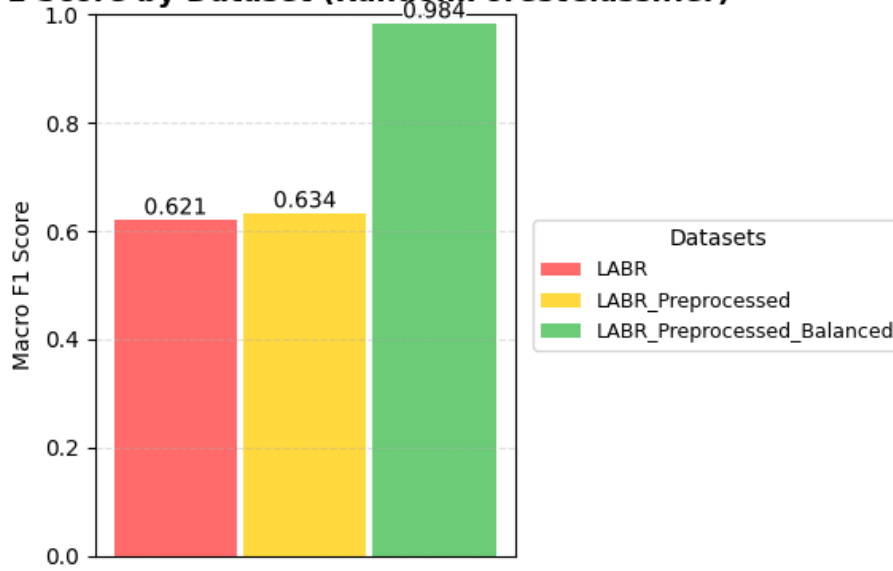
Execution Time (s): 1037.8763

Positive Samples: 40514

Negative Samples: 40514

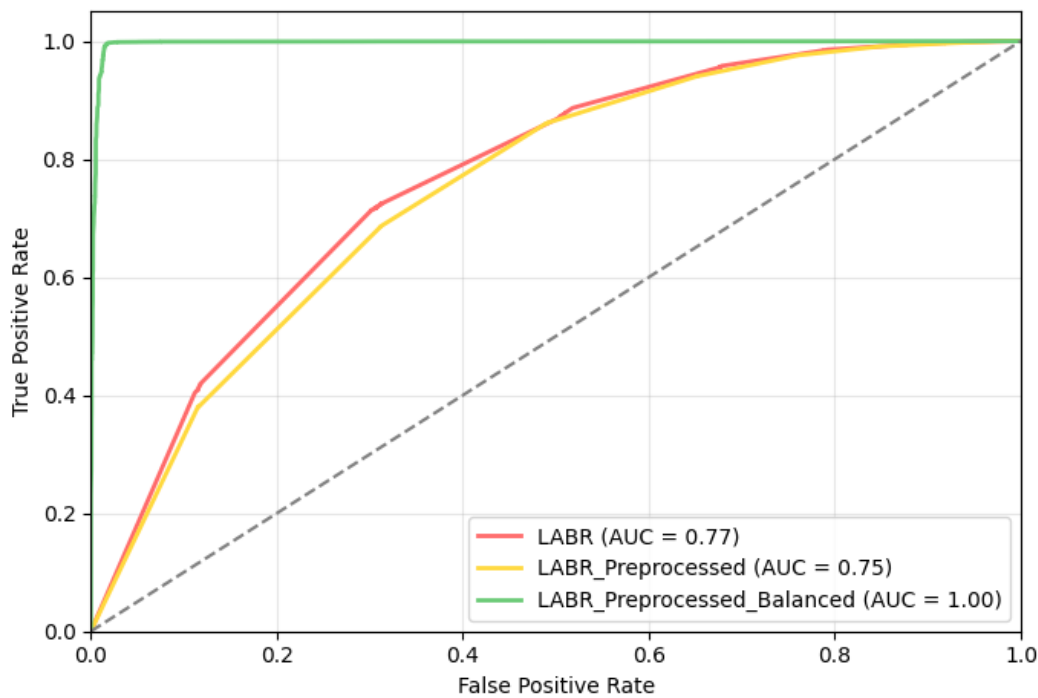
Comparison: Macro F1

Macro F1 Score by Dataset (RandomForestClassifier)



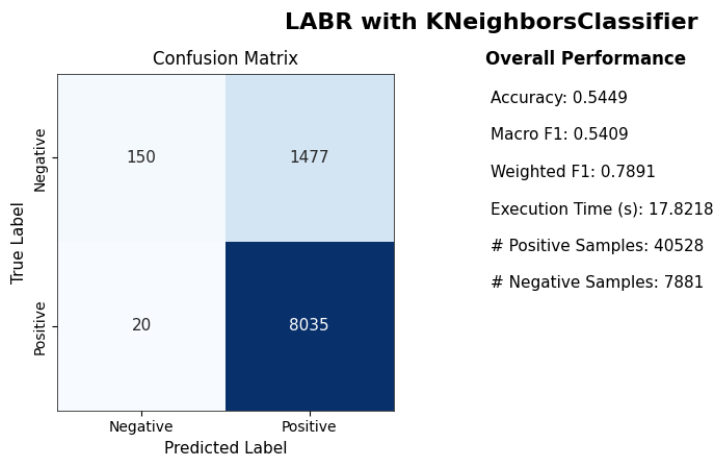
Comparison: ROC

ROC Curves - RandomForestClassifier

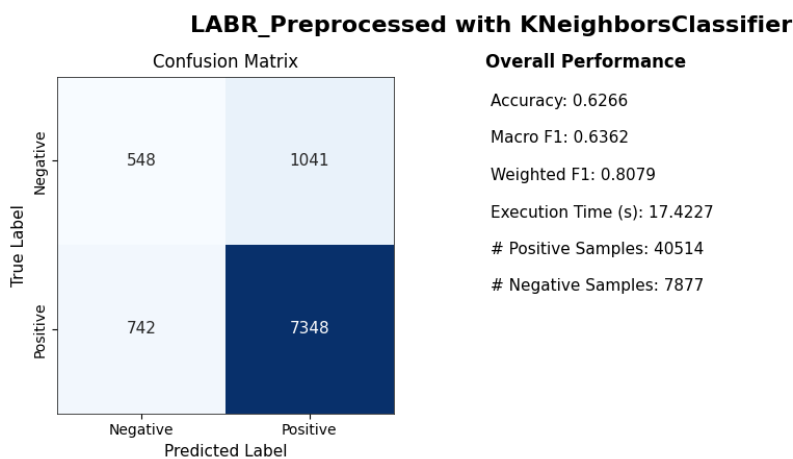


K-Nearest Neighbors

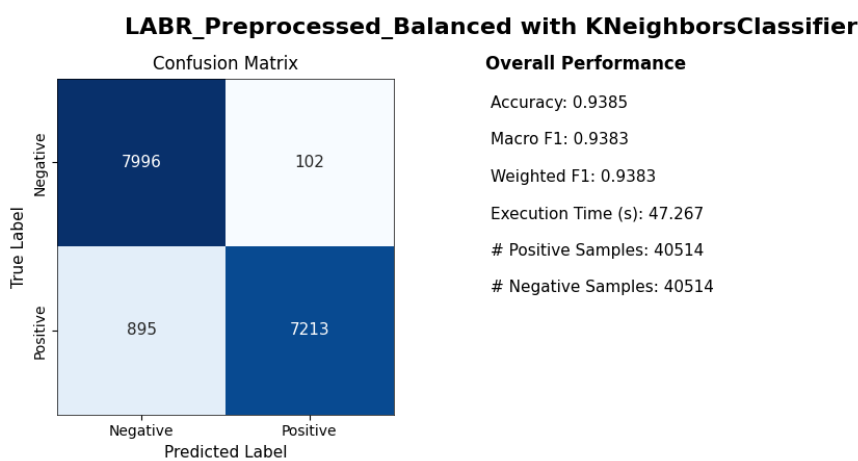
LABR (Brut)



LABR (Prétraité)

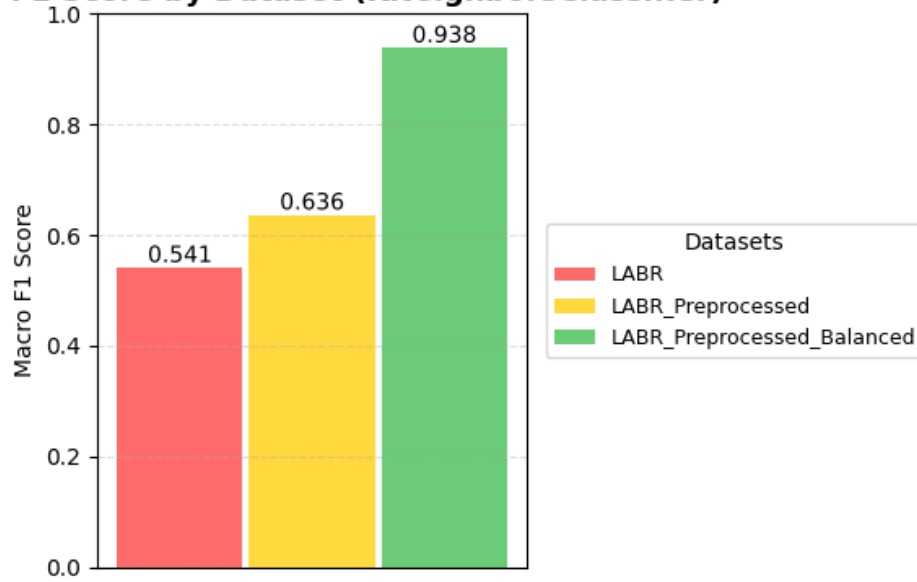


LABR (Prétraité + Équilibré)



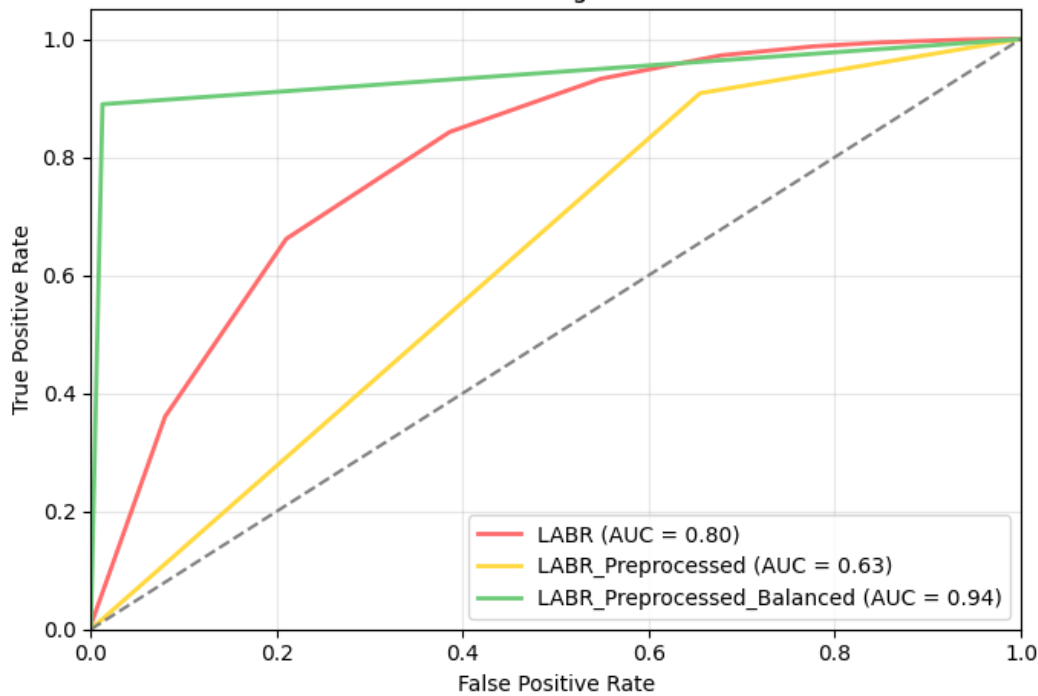
Comparison: Macro F1

Macro F1 Score by Dataset (KNeighborsClassifier)



Comparison: ROC

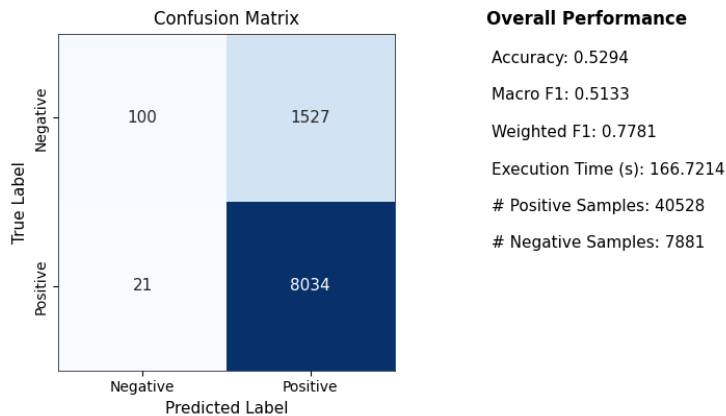
ROC Curves - KNeighborsClassifier



AdaBoost Classifier

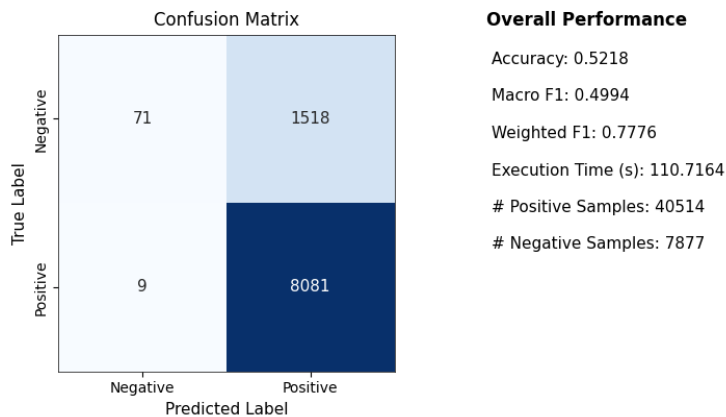
LABR (Brut)

LABR with AdaBoostClassifier



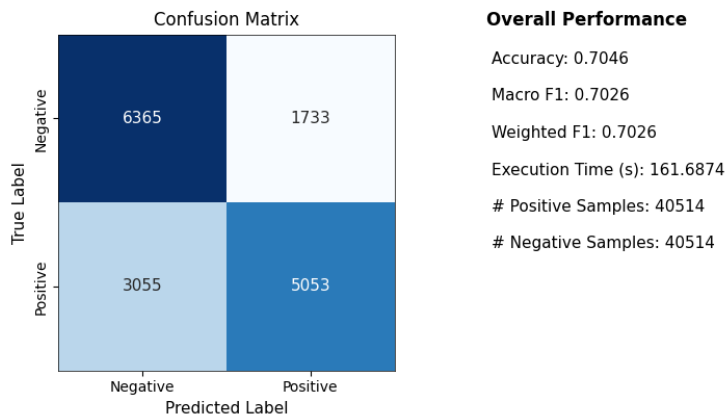
LABR (Prétraité)

LABR_Preprocessed with AdaBoostClassifier



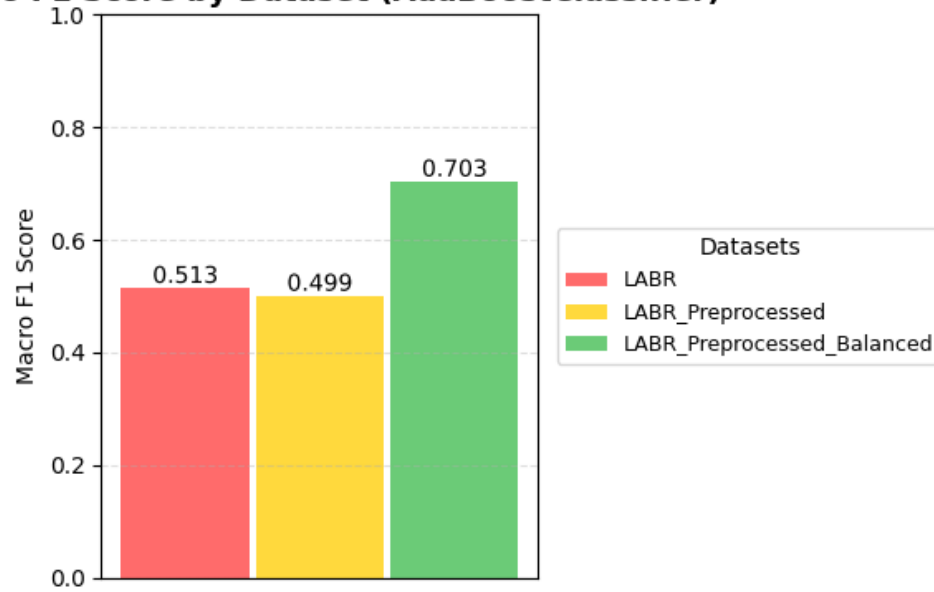
LABR (Prétraité + Équilibré)

LABR_Preprocessed_Balanced with AdaBoostClassifier



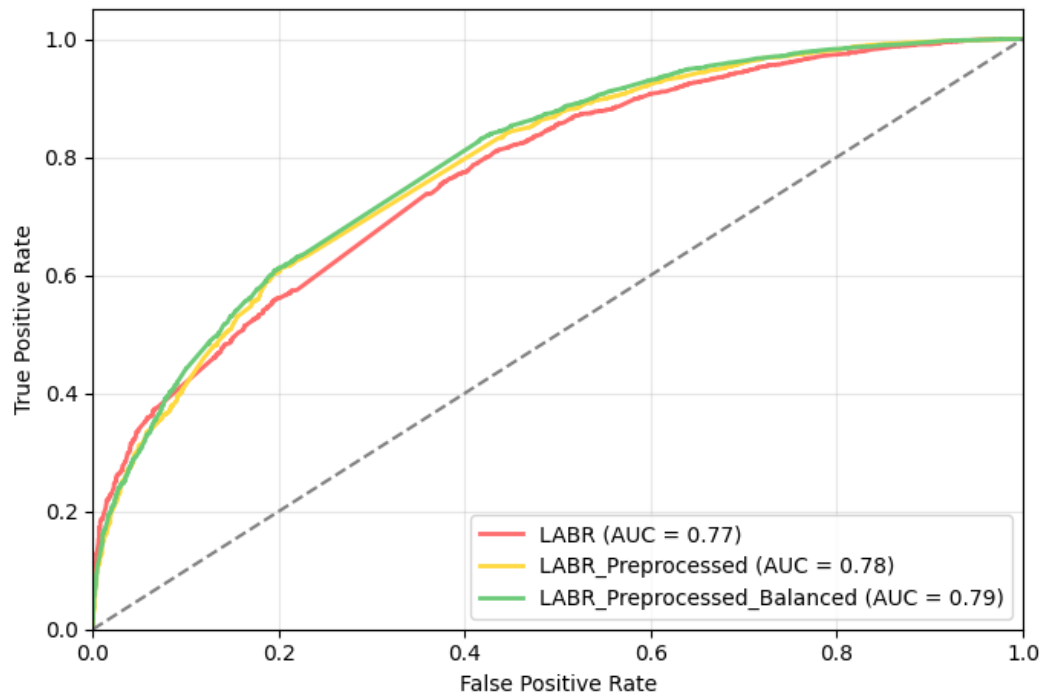
Comparison: Macro F1

Macro F1 Score by Dataset (AdaBoostClassifier)



Comparison: ROC

ROC Curves - AdaBoostClassifier



Global Macro F1 Comparison Across All Models

